

Logistic Regression: regularization (scored task: 4 points)

Tasks (Lab 7):

Recommended functions and packages: `glmnet` (library `glmnet` in R and Python), `LogisticRegression` (library: `scikit-learn` in Python).

1. Logistic regression with different regularizations: lasso, ridge, elastic net.

Consider high-dimensional dataset **prostate** corresponding to predicting the occurrence of prostate cancer using gene expression data. There are 102 observations (patients) and 6033 variables (genes). The response variable indicates whether a patient suffers from disease (prostate cancer, value: 1) or not (value: 0).

- Fit logistic regression models with different regularizations (lasso, ridge, elastic net). Print the values of parameters for different values of parameter λ .
- Make profile plots showing how the values of parameters depend on λ for different variables.
- Select the optimal value of λ using cross-validation.
- Compare the three regularization methods in terms of sparsity, shrinkage, and interpretability of the resulting model.
- Which regularization method sets coefficients exactly to zero? Which one tends to keep correlated variables together? Comment on the observed coefficient paths.
- Does the value of λ selected by cross-validation necessarily correspond to the best model for variable selection? Explain.

2. Performance of lasso for simulated data.

- Generate data from logistic model:

$$y_i \sim \text{Bern}(p_i),$$

where

$$p_i = \frac{1}{1 + \exp[-(\beta^T x_i)]},$$

for $i = 1, \dots, n$, $x_i \sim N(0, I)$ and

$$\beta = (1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0),$$

so there are 10 relevant variables and 10 irrelevant variables.

- Fit logistic model using lasso regularization (use `glmnet` package in R or Python). Use cross-validation to select the optimal value of parameter λ . Report which variables are selected as relevant by the method.

- Compute PSR (Positive Selection Rate)

$$PSR := \frac{|t \cap \hat{t}|}{|\hat{t}|}$$

and FDR (False Discovery Rate)

$$FDR := \frac{|\hat{t} \setminus t|}{|\hat{t}|},$$

where t is a set of true relevant variables and $\hat{t}$ is a set of variables selected by the model as relevant.

- Repeat the above steps $L = 100$ times and compute the averaged values of PSR and FDR.
- Investigate how the averaged values of PSR and FDR depend on the sample size (consider $n = 50, 100, 300, 500, 1000, 2000$); make a plot showing the dependence of PSR and FDR on n .
- Investigate how the averaged values of PSR and FDR depend on the number of irrelevant variables (consider values: 10, 50, 100, 200, 500) for fixed sample size $n = 300$.
- Can a model have good predictive performance and still poor variable selection quality? Compare the roles of PSR and FDR and explain what aspect of performance each of them measures.
- Analyze the robustness of the logistic regression model: generate y_i using

$$p_i = \Phi(\beta^T x_i),$$

where Φ is the distribution function of the standard Gaussian distribution. Repeat the remaining steps.

- Compare the results obtained for logistic and probit-type data generation. Does the lasso procedure still identify the important variables well? Which aspect seems more robust: prediction quality, PSR, or FDR?